

|                         |              | Local Accuracy $\alpha$ |       |       |       | Global Accuracy $\beta$ |       |       |       | Class Separation $\gamma$ |       |       |       |
|-------------------------|--------------|-------------------------|-------|-------|-------|-------------------------|-------|-------|-------|---------------------------|-------|-------|-------|
|                         |              | SOM                     | MDS   | t-SNE | UMAP  | SOM                     | MDS   | t-SNE | UMAP  | SOM                       | MDS   | t-SNE | UMAP  |
| $C_V$                   | LSI          | -0.08                   | 0.29  | 0.10  | -0.11 | 0.07                    | 0.12  | 0.32  | -0.03 | -0.03                     | 0.13  | 0.03  | -0.07 |
|                         | LSI & tf-idf | 0.28                    | 0.21  | 0.42  | 0.18  | 0.26                    | -0.10 | 0.11  | 0.18  | 0.33                      | 0.15  | 0.23  | 0.25  |
|                         | NMF          | 0.01                    | 0.41  | 0.00  | -0.05 | 0.00                    | 0.00  | 0.00  | -0.06 | -0.25                     | 0.12  | -0.20 | -0.27 |
|                         | NMF & tf-idf | 0.20                    | 0.12  | 0.38  | 0.13  | 0.01                    | -0.01 | -0.05 | 0.00  | -0.07                     | 0.01  | -0.12 | -0.09 |
|                         | LDA          | 0.02                    | -0.03 | 0.01  | -0.05 | 0.02                    | 0.26  | 0.04  | -0.04 | -0.35                     | -0.28 | -0.28 | -0.32 |
| $U_{\text{Mass}}$       | LSI          | -0.15                   | 0.15  | 0.15  | -0.18 | 0.06                    | 0.19  | 0.29  | -0.06 | 0.06                      | 0.03  | -0.03 | 0.04  |
|                         | LSI & tf-idf | -0.47                   | 0.28  | 0.00  | -0.43 | 0.22                    | 0.24  | 0.54  | 0.25  | -0.11                     | 0.30  | -0.27 | -0.08 |
|                         | NMF          | 0.16                    | -0.08 | 0.16  | 0.13  | 0.22                    | 0.26  | 0.15  | 0.15  | 0.13                      | 0.01  | 0.06  | 0.12  |
|                         | NMF & tf-idf | 0.11                    | 0.45  | 0.23  | 0.03  | 0.28                    | 0.25  | 0.20  | 0.18  | -0.30                     | 0.19  | -0.39 | -0.32 |
|                         | LDA          | 0.02                    | 0.22  | -0.06 | 0.00  | 0.10                    | 0.06  | -0.18 | -0.02 | 0.02                      | 0.32  | -0.12 | -0.05 |
| $C_{\text{UCI}}$        | LSI          | -0.17                   | 0.37  | 0.14  | -0.18 | 0.03                    | 0.38  | 0.24  | -0.07 | -0.16                     | -0.06 | -0.28 | -0.17 |
|                         | LSI & tf-idf | -0.43                   | 0.29  | -0.01 | -0.38 | 0.21                    | 0.40  | 0.46  | 0.24  | -0.27                     | 0.04  | -0.40 | -0.23 |
|                         | NMF          | 0.13                    | 0.50  | 0.20  | 0.04  | 0.20                    | 0.38  | 0.05  | 0.03  | -0.39                     | -0.05 | -0.43 | -0.40 |
|                         | NMF & tf-idf | 0.13                    | 0.51  | 0.26  | 0.06  | 0.26                    | 0.26  | 0.12  | 0.15  | -0.44                     | 0.05  | -0.55 | -0.46 |
|                         | LDA          | 0.06                    | 0.20  | -0.04 | -0.03 | 0.22                    | 0.41  | -0.18 | -0.05 | -0.23                     | -0.03 | -0.38 | -0.31 |
| $C_{\text{NPMI}}$       | LSI          | -0.19                   | 0.29  | 0.10  | -0.21 | 0.05                    | 0.21  | 0.35  | -0.06 | -0.05                     | 0.12  | -0.09 | -0.08 |
|                         | LSI & tf-idf | -0.22                   | 0.44  | 0.19  | -0.22 | 0.33                    | 0.51  | 0.48  | 0.34  | -0.17                     | -0.12 | -0.34 | -0.15 |
|                         | NMF          | 0.03                    | 0.47  | 0.02  | -0.05 | 0.08                    | 0.02  | 0.05  | -0.01 | -0.37                     | 0.17  | -0.35 | -0.39 |
|                         | NMF & tf-idf | 0.18                    | 0.51  | 0.37  | 0.13  | 0.27                    | 0.14  | 0.11  | 0.18  | -0.55                     | -0.15 | -0.67 | -0.58 |
|                         | LDA          | 0.03                    | 0.04  | 0.00  | -0.07 | 0.17                    | 0.63  | -0.05 | -0.02 | -0.53                     | -0.39 | -0.64 | -0.61 |
| BERT Sentence Embedding |              | 0.18                    | 0.19  | 0.26  | 0.22  | 0.27                    | -0.47 | 0.31  | 0.30  | 0.08                      | 0.03  | 0.09  | 0.13  |
| BERT Semantic Search    |              | 0.18                    | 0.18  | 0.27  | 0.21  | 0.27                    | -0.47 | 0.33  | 0.31  | 0.08                      | 0.01  | 0.09  | 0.13  |